ANTICIPY

Manufacturing Brief

Project: Anticipy Pendant

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1. The Ask

We are adapting an open-source, MIT-licensed wearable hardware reference into a production product. We are not asking you to design from scratch. We are asking you to take a proven reference design on the exact silicon we need, fit it into our titanium industrial design, and deliver prototype units followed by a production run.

This is one engagement, with one commercial number for prototype delivery and one for production delivery. We are evaluating two vendors in parallel and would like your response within seven calendar days of receiving this brief and the accompanying enclosure STEP file.

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2. The Product

Anticipy is a small wearable pendant that continuously captures ambient audio and streams it via Bluetooth Low Energy to a paired phone, which forwards the audio to the user's private server. The pendant performs no on-device AI, no transcription, and no decision-making. It is a microphone and a radio. All intelligence runs downstream of the wearable, off the device entirely.

Three implications of that for engineering. The pendant needs no application processor headroom beyond the BLE SoC. Audio buffering must be robust because the paired phone is intermittently out of range. And the BLE link is the single most important runtime characteristic, since a disconnected pendant is a non-functional product.

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3. Reference Design

The starting point is the Omi Consumer Version 1 (CV1), published by Based Hardware under the MIT license. Same silicon we target, dual microphones, BLE, and the full design package is public: PCB schematics, Gerbers, Altium source, mechanical STEP, BOM, and assembly guides.

Reference	Omi Consumer Version 1 (CV1) by Based Hardware
License	MIT (commercial use, modification, and closed derivatives permitted)
Repository	github.com/BasedHardware/omi
Documentation	docs.omi.me
Primary SoC	Nordic Semiconductor nRF5340 (dual-core Arm Cortex-M33, BLE 5.x)
Wi-Fi co-processor	Nordic nRF7002 (Wi-Fi 6) — see note in Section 5
Microphones	Dual TDK InvenSense T5838 top-port PDM
Battery / charging	Per Omi reference; we are open to vendor adaptation

Where this brief is silent, treat the Omi reference as the default. Where this brief asks for a change, the change takes precedence.

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4. Industrial Design and Mechanical

Enclosure	CNC-machined Grade 5 titanium (Ti-6Al-4V), brushed finish, hairline grain along long axis
Construction	Two-piece case with single hairline parting line, no visible fasteners on user-facing surfaces
Target weight (assembled)	Under 15 grams, with under 10 grams as a stretch goal
Ingress protection	IP67 (dust-tight, 30 minutes at 1 m immersion). Not a swim product.
Charging interface	Magnetic, rear face. Pogo pins is the default; vendor may propose wireless Qi if shell volume permits.
Wear interface	Pendant chain loop and magnetic clasp face. Geometry per supplied STEP.
Marking	Single laser-etched mark on rear, placement open to vendor recommendation

We will provide the current titanium enclosure STEP file under separate cover. Internal dimensions are fixed. Surface finish details, parting line, and laser etch placement are open for your input.

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5. Electronics: What Changes From the Reference

Three electronics decisions are open for your engineering input. Everything else should follow the Omi CV1 reference.

- **Wi-Fi co-processor.** The Omi CV1 includes the nRF7002 for Wi-Fi 6. Anticipy is phone-tethered, so the pendant does not require independent internet connectivity. Please advise whether removing the nRF7002 is recommended to reduce BOM cost, power draw, and board area, or whether keeping it open for future use cases is worth the cost.
- **Battery and runtime.** Target runtime is one full day of continuous capture, approximately 16 hours, with BLE active. Please specify the battery capacity, chemistry, and physical dimensions that achieve this within the shell volume we provide, and confirm whether the Omi reference battery selection is appropriate or should be changed.
- **Onboard buffer storage.** When the paired phone is out of range, the pendant must buffer captured audio locally and deliver it on reconnection. Please specify the onboard flash capacity you would recommend, with an absolute minimum of eight hours of buffered compressed audio.

Microphones should be the dual T5838 PDM mics as in the reference, unless you have a strong reason to substitute.

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6. Antenna (Critical)

This is the single most important technical question in the brief, and the section that must be answered explicitly in writing in your quote.

The Omi reference has a BLE antenna tuned for its plastic enclosure. Anticipy uses a titanium enclosure. Titanium is conductive and will detune, attenuate, or fully block a standard antenna layout designed for plastic. A direct port of the reference antenna into a metal case will not yield a functional BLE link.

The proven industry solution, used by Apple Watch and similar premium titanium wearables, is an antenna window: a non-conductive insert (engineering plastic, ceramic, or sapphire) integrated into the metal enclosure, with a tuned chip antenna positioned behind it and a matching network sized to the new structure. This is the approach we expect you to follow unless you have a specific reason to propose otherwise.

Required in the quote response:

- Confirmation that you can design and tune a BLE antenna for a titanium enclosure using an antenna-window approach (or your alternative).

- Name of the RF engineer responsible.
- Confirmation that electromagnetic simulation in the actual enclosure geometry, plus bench-measured antenna performance on the first prototype, will be delivered as part of the engagement.

If this work cannot be done in-house, please say so plainly. Subcontracting is acceptable, provided the subcontractor is named and the work is covered under your warranty.

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7. Firmware

The starting point is Omi's open-source nRF5340 firmware, MIT licensed, in the same repository as the hardware. It handles dual-mic capture, BLE streaming, local buffering, and OTA updates. Adapting it to whatever board you produce is the only firmware work in question.

Please indicate in your quote which of the following applies, and price your turnkey numbers accordingly:

- **Option A.** You deliver bring-up and production firmware on the adapted hardware, including OTA infrastructure. Quote your turnkey numbers with firmware fully included.
- **Option B.** You deliver bring-up firmware sufficient to validate the adapted hardware, and provide interface documentation. We complete the production firmware. Quote your turnkey numbers with bring-up only included.
- **Option C.** Hardware only. We take full firmware responsibility from the start, against the schematics and pinout you provide. Quote your turnkey numbers with no firmware in scope.

All three options are acceptable. We just need to know which one your quote reflects.

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8. Quality and Certifications

Standard consumer electronics quality, not medical-device quality. ISO 9001 facility certification is preferred. ISO 13485 is not required for this product.

On every shipped unit: BLE pairing test, audio capture verification on both microphones, full charge cycle, and serial number laser-etched on the rear. Sample-basis testing for IP67 immersion and one-meter drop on each production batch. Per-unit test results retained by your facility for twelve months.

Certifications required for launch:

- FCC Part 15 Subpart B (United States)
- ISED RSS-247 (Canada)
- Bluetooth SIG QDID declaration

CE RED for the European Union and UKCA for the United Kingdom are phase 2, not required for the initial production run. Please note in your response which certifications are covered by your turnkey production number and which would be billed separately if they fall outside it.

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9. What We Want Delivered

- **Adapted hardware files.** Final Gerber set, updated BOM with current sourcing and stock status, and updated mechanical STEP showing PCB-to-enclosure fit.
- **Antenna design report.** Simulation plus measured performance on the first prototype, per Section 6.
- **Prototype batch.** Five to ten fully assembled and tested units shipped DDP to Vancouver, Canada.
- **Production-ready files.** Full release package required to run the first one thousand-unit batch with no further engineering changes.
- **Production batch.** One thousand fully assembled, tested, and serialized units shipped DDP to Vancouver.
- **Firmware artifacts.** Per the option selected in Section 7.
- **Design History File.** Engineering decisions, simulations, test reports, and quality records collected at engagement close.

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10. What We Will Provide

- Titanium enclosure STEP file, current revision. Internal dimensions fixed; finish and parting line open.
- Direct technical access to the founder for any engineering questions during the engagement, with a target response time under one business day.
- BLE staging endpoint and a test phone configured to receive the audio stream, for prototype validation.
- Prompt review of vendor questions, design proposals, and simulations.

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11. Quote Scope

Two turnkey numbers, all-inclusive. Please do not return an itemized list of design fees, NRE, antenna fees, certification fees, tooling fees, or per-unit costs as separate line items. Two numbers is all we need to compare quotes.

Turnkey number A	All-in cost to deliver a prototype batch of 5 to 10 fully assembled

	and tested units in Vancouver, Canada, including all engineering, antenna work, and firmware per Section 7. One number.
Turnkey number B	Per-unit cost at 1,000 units, all-in, DDP Vancouver, including required certifications per Section 8. One number.
Directional only	Per-unit cost projection at 5,000 and 10,000 units, non-binding. One number each.

If something is not included in your turnkey numbers, please say what it is and price it separately at the bottom. The default assumption is that the turnkey numbers cover everything required to put working, certified units in our hand.

Payment terms: we prefer milestone-based payments tied to deliverables. Please propose a payment schedule. Currency in USD.

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12. Timeline

Please specify working-day estimates from receipt of signed engagement letter and STEP file to each milestone below:

- Design and antenna work complete, files delivered.
- First prototype units in hand in Vancouver.
- Production-ready release, all certifications complete.
- First 1,000-unit batch shipped DDP.

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13. Communication

Primary contact	Omar Ebrahim, Founder, Anticipy
Email	hello@anticipy.ai
Time zone	Pacific Time (Vancouver, Canada), UTC minus 7
Real-time chat	WeChat or WhatsApp on vendor preference
Quote due	Within 7 calendar days of receiving this brief and the STEP file
Post-call summary	Written scope agreement within 24 hours of our kickoff call

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14. Reference Quick Links

Omi repository	github.com/BasedHardware/omi
Omi consumer hardware documentation	docs.omi.me/doc/hardware/OmiConsumer
Open-source hardware design files	docs.omi.me/doc/hardware/consumer
License (MIT)	Commercial use, modification, and closed derivatives permitted with attribution
Nordic nRF5340 datasheet	nordicsemi.com/Products/nRF5340
TDK InvenSense T5838 microphone	invensense.tdk.com (search T5838)

Please confirm receipt of this brief and the accompanying STEP file. Two turnkey numbers within seven calendar days. Thank you.